

Model vs Index — CAGR Comparison

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What period this covers

Item	Value
Test window	2020-07-01 to 2025-01-31
Rebalance frequency	monthly
Rebalance dates	51
Elapsed time	4.25 years
Holding period	entry to the next rebalance's entry (chaining)
Portfolio	equal-weighted top 20, rebalanced each period
CAGR formula	product of (1 + period return) over 51 periods, then raised to 1/4.25

Basis

Price return compared with price return. NSE's historical index endpoint serves price indices only — every total-return (TRI) spelling returns no data. The strategy's dividends are therefore reported separately rather than folded into the headline, because comparing a dividend-inclusive strategy against a dividend-less index would credit the model with roughly the market's yield (about 1–1.5% a year in India) for nothing.

Index returns are measured over the sessions the strategy actually traded — the same next-session entry and the same exit — not between month-ends, so neither side is measuring a different span of market time.

Headline: CAGR by strategy

Strategy	Gross	Net	vs UNIVERSE	vs N50	vs N500	vs N500vs NMIDCAP 150	vs N500vs NSMALLCAP 250
equal weight universe	44.13%	42.61%	-1.52%	+22.94%	+19.78%	+9.84%	+7.53%
growth only	50.16%	46.61%	+2.48%	+26.95%	+23.79%	+13.85%	+11.53%
model 5	70.44%	61.02%	+16.89%	+41.35%	+38.19%	+28.25%	+25.94%
momentum only	57.50%	42.91%	-1.22%	+23.24%	+20.08%	+10.14%	+7.83%
quality only	34.67%	32.82%	-11.31%	+13.15%	+9.99%	+0.05%	-2.26%
random ranking	42.94%	17.44%	-26.69%	-2.23%	-5.39%	-15.33%	-17.64%
simple quality momentum	39.90%	31.17%	-12.96%	+11.50%	+8.34%	-1.60%	-3.91%
value only	101.01%	93.23%	+49.10%	+73.56%	+70.40%	+60.46%	+58.15%

vs UNIVERSE is the column that matters, and it is deliberately placed before the index columns. It compares the strategy against an equal-weighted holding of the very same point-in-time universe it selected from, so it isolates stock selection.

Why the index columns flatter every strategy. The eligible universe returned 44.13% against NIFTY 500's 22.83% — a +21.30 point gap earned purely by equal-weighting a broad, small-cap-heavy universe, before any selection at all. That gap is inside every "vs index" number below. A strategy that beats NIFTY 500 by 20 points while matching the universe has selected nothing.

Index reference

Index	CAGR	Volatility	Sharpe	Max drawdown	Periods
NIFTY 50	19.67%	19.04%	1.03	-12.15%	51
NIFTY 500	22.83%	20.10%	1.12	-12.47%	51
NIFTY MIDCAP 150	32.77%	24.37%	1.28	-13.67%	51
NIFTY SMALLCAP 250	35.08%	28.30%	1.20	-18.67%	51

Eligible universe (equal weight)	44.13%	-	-	-	51
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The **eligible universe** row is the benchmark that controls for size and breadth: the same point-in-time universe the strategy chose from, held equally weighted. Beating NIFTY 500 while trailing this means the model captured a small-cap tilt rather than stock selection.

Risk-adjusted comparison

Strategy	Basis	CAGR	Volatility	Sharpe	Max DD	Hit rate
equal weight universe	net	42.61%	29.13%	1.36	-15.24%	67%
growth only	net	46.61%	29.80%	1.44	-23.71%	69%
model 5	net	61.02%	28.91%	1.81	-15.22%	71%
momentum only	net	42.91%	35.00%	1.20	-34.51%	65%
quality only	net	32.82%	22.99%	1.35	-24.47%	67%
random ranking	net	17.44%	33.11%	0.63	-33.68%	55%
simple quality momentum	net	31.17%	28.21%	1.11	-26.51%	67%
value only	net	93.23%	41.00%	1.82	-19.71%	71%

Excess return against NIFTY 500

Strategy	Mean excess/period	Tracking error	Information ratio	Periods beaten
equal weight universe	+1.43%	15.92%	1.08	63%
growth only	+1.70%	20.12%	1.01	57%
model 5	+2.48%	19.60%	1.52	63%
momentum only	+1.62%	26.22%	0.74	59%
quality only	+0.72%	13.70%	0.63	51%
random ranking	-0.13%	20.04%	-0.08	47%
simple quality momentum	+0.73%	20.65%	0.42	55%
value only	+4.34%	28.51%	1.83	65%

Information ratio is excess return per unit of tracking error — how reliably the outperformance was earned, not how large it was. **Periods beaten** is the share of rebalances where the strategy outpaced the index; a high CAGR with a low share was carried by a handful of periods.

Turnover and cost drag

Strategy	One-way turnover	Gross CAGR	Net CAGR	Cost drag
equal weight universe	0.055	44.13%	42.61%	1.52%
growth only	0.122	50.16%	46.61%	3.54%
model 5	0.361	70.44%	61.02%	9.43%
momentum only	0.486	57.50%	42.91%	14.59%
quality only	0.086	34.67%	32.82%	1.86%
random ranking	0.973	42.94%	17.44%	25.50%
simple quality momentum	0.419	39.90%	31.17%	8.73%
value only	0.173	101.01%	93.23%	7.78%

Cost drag is annualised, and it is the difference between a ranking that works on paper and one that works in an account. A strategy replacing most of its book every month pays it repeatedly.

Before quoting these numbers

An index is investable and this portfolio is a simulation. The index CAGR is achievable through a low-cost fund; the strategy CAGR assumes every fill happened at the modelled price and size.

The window is what it is. A CAGR over a few years is dominated by the market regime inside it. Check the periods-beaten share and the drawdown before treating a CAGR gap as skill.

Costs are modelled, not measured. Statutory charges are exact and effective-dated; half-spread and market impact are assumptions, which is why gross and net are both shown.

Delisting recovery is an assumption. Re-run with --delisting-strategy zero and last_close to see how much of the gap depends on it.